Semantic Networks, Frames, and Ontologies in Knowledge, Reasoning, and Learning

- **Topic:** Semantic Networks, Frames, and Ontologies
- **Subtitle:** Techniques in Knowledge Representation & Reasoning

Introduction to Knowledge Representation (KR)

- Why KR is essential in AI:
- Representing the world
- Reasoning about data
- Learning from experience
- Key KR techniques:
- Semantic Networks
- Frames
- Ontologies

Semantic Networks – Overview

- Graph-based structure representing knowledge
- **Nodes** = Concepts/Entities
- **Edges** = Relationships
- Example: “A cat is a mammal”

  → Cat – *is-a* → Mammal

Types of Semantic Networks

- **Definitional networks**: Taxonomies
- **Assertional networks**: Facts and instances
- **Implication networks**: Rule-based
- **Executable networks**: Include procedures

Slide 5: Semantic Network Example

Visual diagram:

- Bird → *is-a* ⟶ Animal
- Bird → *can* ⟶ Fly
- Penguin → *is-a* ⟶ Bird
- Penguin → *cannot* ⟶ Fly

Inference in Semantic Networks

- Property inheritance
- Hierarchical reasoning
- Forward and backward chaining
- Example:
- All mammals breathe → Dog is a mammal → Dog breathes

Frames – Introduction

- Structuring knowledge in “data slots”
- Frame = Object + Attributes
- Each attribute (slot) may contain:
  - Default values
  - Procedural attachments (if-needed, if-added)
- Inspired by object-oriented concepts

Frame Structure Example

- **Frame: Car**
- Type: Vehicle
- Engine: Petrol
- Wheels: 4
- Owner: Person
- Example: Toyota-Camry → instance of Car

Reasoning with Frames

- Inheritance of slot values
- Slot constraints and conditions
- Procedural reasoning via slot triggers
- Example: If Door is open, trigger alarm

Ontologies – Definition

- Formal, structured set of terms and relationships
- Used to model domains of knowledge
- Define:
  - **Concepts (Classes)**
  - **Attributes (Properties)**
  - **Relations (Object/Data properties)**
  - **Individuals (Instances)**

Components of Ontologies

- **Classes** – Abstract groups (e.g., Person, Animal)
- **Properties** – Describe attributes or relations
- **Instances** – Specific objects (e.g., "Alice" is a Person)
- **Axioms** – Rules and constraints
- **Hierarchy** – SubClassOf, EquivalentClass

Example Ontology (Healthcare Domain)

- **Class**: Patient, Doctor, Disease
- **Object Property**: treats, suffersFrom
- **Data Property**: hasAge, hasName
- Visual diagram of a simple ontology (can be made in Protégé)

Ontology Languages and Tools

- **Languages:**
- OWL (Web Ontology Language)
- RDF (Resource Description Framework)
- RDFS (RDF Schema)
- **Tools:**
- Protégé
- OntoGraf
- TopBraid Composer

Applications in AI

- **Semantic Search**
- **Knowledge Graphs (e.g., Google KG)**
- **Chatbots / Virtual Assistants**
- **Natural Language Understanding**
- **Recommendation Systems**
- **Medical Diagnosis** (ontology-driven)

Comparison Table

Feature	Semantic Networks	Frames	Ontologies
Structure	Graph	Slot-based	Class-based
Inference	Inheritance	Procedural	Logical
Tools	Custom, NLTK	Custom, CLIPS	OWL, Protégé
Domain Use	General	Specific	Structured domains

Challenges

- Ambiguity in semantic definitions
- Integration of different KR models
- Maintenance and evolution of ontologies
- Scalability in large systems

Future Directions

- **Neuro-symbolic AI**: Combining deep learning with symbolic KR
- **Ontology Learning**: Automatically creating ontologies from data
- **Linked Data and Semantic Web**
- **Integration with Reinforcement Learning**